

Assigned: 02nd January 2025

Due: 17th January 2025 before 11.59PM

Marks: 20 marks possible

Submission Procedure:

- **Create / debug / test your assignment in Visual Studio**
- Create a **ZIP** file of your **entire CareerCloud solution** and upload it to **BlackBoard**.
 - o Please ensure you upload the entire solution.
 - o Name your ZIP file according to the guidelines in the '1 – Sundry' slide deck
 - E.g. NET.22-4-Joe-Smith-N#####.zip
- **DO NOT EMAIL YOUR ASSIGNMENT** – The Humber email server will not deliver the assignment and you will not be notified that it was not delivered.

Description:

In this assignment, we'll implement an Entity Framework data layer. The Entity Framework layer will implement the IRepository interface which means that we can interchange this layer for the ADO data layer we constructed in NET .12.

Steps:

1. Add navigation properties to Poco classes.
 - a. The database foreign keys are represented by creating a property that references the type at the opposite end of the relationship. Relationships (one-to-many) that reference many items on the opposite end are represented by an ICollection<T>.
 - i. For example, 'CompanyDescriptionPoco' will reference a single 'CompanyProfilePoco' record and 'CompanyProfilePoco' will reference multiple 'CompanyDescriptionPoco' records.
 1. The navigation property in the 'CompanyDescriptionPoco' class to 'CompanyProfilePoco' will be:
 - a. `public virtual CompanyProfilePoco CompanyProfiles { get; set; }`
 2. The navigation property in the 'CompanyProfilePoco' class to 'CompanyDescriptionPoco' will be:
 - a. `public virtual ICollection<CompanyDescriptionPoco> CompanyDescriptions { get; set; }`
2. Add a new class library project to the 'CareerCloud' solution called 'CareerCloud.EntityFrameworkDataAccess'. **The framework selected should be .NET 8.0 (Long Term Support).**
3. Add a reference in 'CareerCloud.EntityFrameworkDataAccess' to the 'Entity Framework Core' NuGet package "install-package Microsoft.entityframeworkcore" (make sure to select the

correct project targeting .NET Core version) and the 'Microsoft.EntityFrameworkCore.SqlServer' package (Install-Package microsoft.entityframeworkcore.SqlServer).

(Ensure that you choose the correct packages compatible with .NET 8)

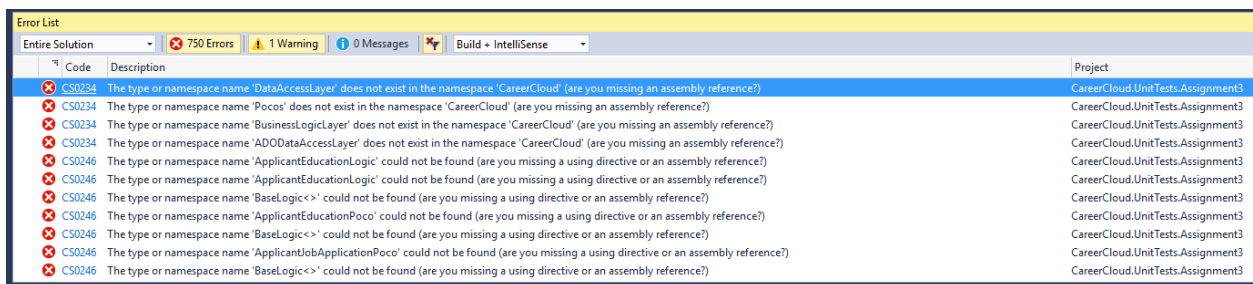
4. Create a class called 'CareerCloudContext' in 'CareerCloud.EntityFrameworkDataAccess'. This class will form the data context implementation for the project and implements the following specifications:
 - a. Inherit from Entity Framework's 'DbContext'
 - i. Override the 'OnConfiguring' method and passes the database connection string to the 'UseSqlServer' method.
 - b. Exposes a 'DbSet' property for each Poco in the 'CareerCloud.Pocos' project.
 - i. The DbSet property will have the same name as the Poco class without the Poco extension. For example: The DbSet for 'CompanyLocationPoco' will be called 'CompanyLocations'
 - c. Add relations to the context class by overriding the 'OnModelCreating' method of the base 'DbContext' class and describing the foreign key relationships to Entity Framework (where required)
5. Create a class called 'EFGenericRepository' in the 'CareerCloud.EntityFrameworkDataAccess' project. This class must follow the following specification:
 - a. Takes a generic parameter
 - b. Constrain the generic parameter to reference types only.
 - c. Inherit from and implement all the required methods as dictated IRepository.

Testing Your Code:

You can test your code before submitting. There is a project on GitHub located here:

<https://github.com/johnhinz/CareerCloudCore.UnitTests.Assignment4.git>

- Clone the solution from GitHub using Visual Studio.
- Open the solution.
- Build the test project
 - You should receive an error like the following (these errors are okay):



- Copy the test project folder into your CareerCloud solution folder.
- Add the project to your CareerCloud solution through Visual Studio
 - In your assignments solution explorer window right click the solution and select 'Add' -> 'Existing project'

- Navigate to the CareerCloud.UnitTests.Assignment4 directory you copied previously
- Add the 'CareerCloud.UnitTests.Assignment4.csproj' to the CareerCloud solution
- Build the CareerCloud solution.
- Open the Test Explorer window. (Menu: Test -> Windows -> Test Explorer)

You should now be able to run the unit tests on the assignment.

Marking Scheme:

Item	Maximum Mark	Your Mark
Add navigation properties to the Poco classes	3	3
Create 'CareerCloud.EntityFrameworkDataAccess' in the 'CareerCloud' solution	1	1
Add a NuGet reference to the 'EntityFramework' in the 'CareerCloud.EntityFrameworkDataAccess' project.	1	1
Creation of Entity Framework Data Context Class	5	5
Implement the 'EFGenericRepository' class. All method as dictated by the IRepository<T> interface.	10	10
Total	20	20

Reference:

Title	Link
Introduction to Entity Framework	https://msdn.microsoft.com/en-us/library/aa937723(v=vs.113).aspx
Entity Framework	https://docs.microsoft.com/en-us/ef/
Entity Framework Fluent API - Relationships	https://msdn.microsoft.com/en-us/library/jj591620(v=vs.113).aspx
Entity Framework Add and Attach and Entity States	https://msdn.microsoft.com/en-us/library/jj592676(v=vs.113).aspx